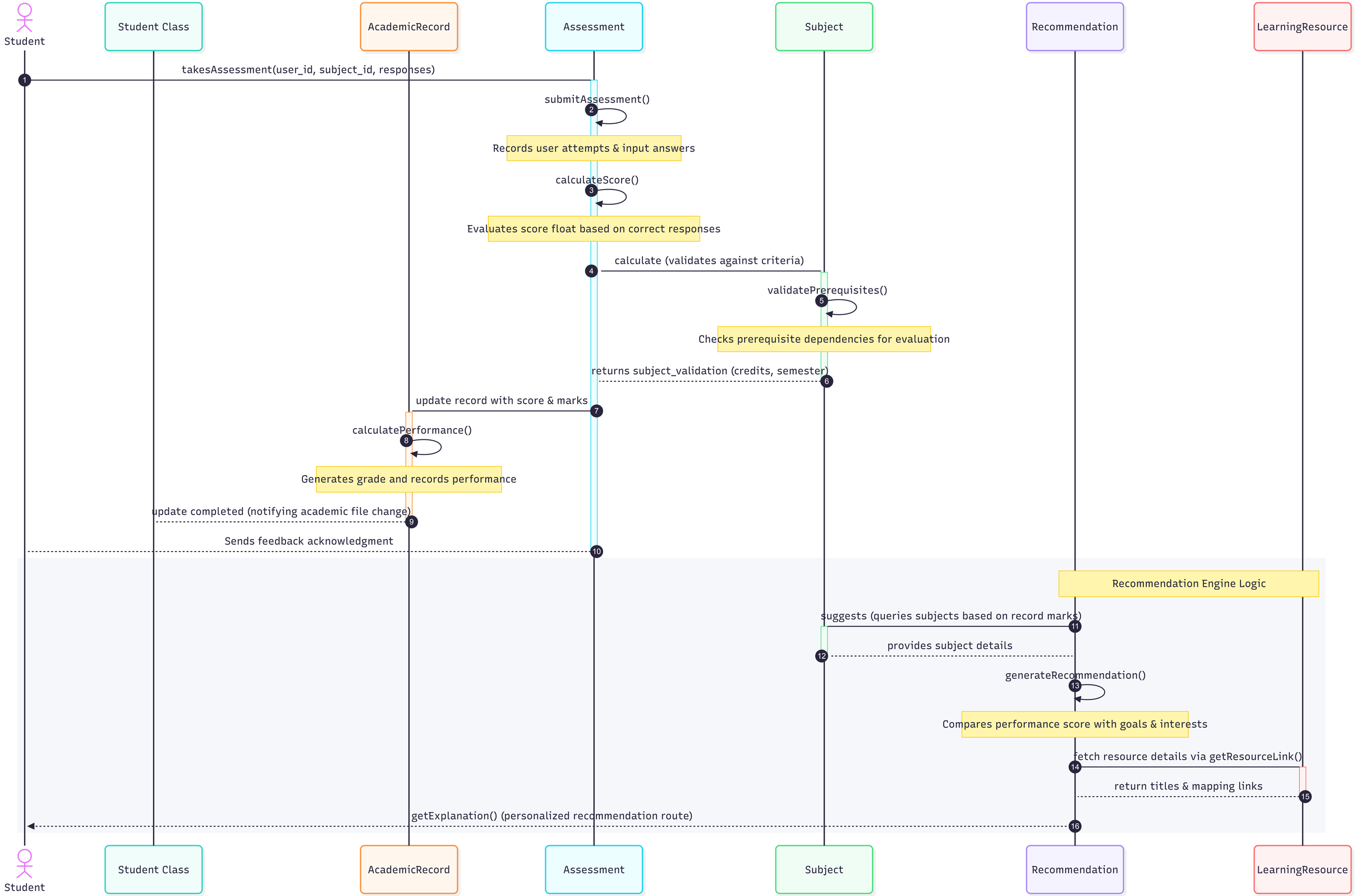

Student
Student Class
AcademicRecord
Assessment
Subject
Recommendation
LearningResource
1
takesAssessment(user_id, subject_id, responses)
submitAssessment()
2
Records user attempts & input answers
calculateScore()
3
Evaluates score float based on correct responses
calculate (validates against criteria)
4
validatePrerequisites()
5
Checks prerequisite dependencies for evaluation
returns subject_validation (credits, semester)
6
update record with score & marks
7
calculatePerformance()
8
Generates grade and records performance
update completed (notifying academic file change)
9
Sends feedback acknowledgment
10
Recommendation Engine Logic
suggests (queries subjects based on record marks)
11
provides subject details
12
generateRecommendation()
13
Compares performance score with goals & interests
fetch resource details via getResourceLink()
14
return titles & mapping links
15
getExplanation() (personalized recommendation route)
16
Student
Student Class
AcademicRecord
Assessment
Subject
Recommendation
LearningResource